





Volumetric Error Compensation in CNC Machine Tools and CMS

TOPIC relevant also for COORD. MEAS. SYSTEM (CMM)
↑
that reasoning can be used to increase cutting accuracy
(increase performance of CMM)

Hocken R.J., P.H. Pereira (2012)
Coordinate Measuring Machines and Systems
Chapter 12



The student knows:

- What the volumetric error of a CMM/CMS/MT is
- How to model the volumetric error
- How to compensate the volumetric error



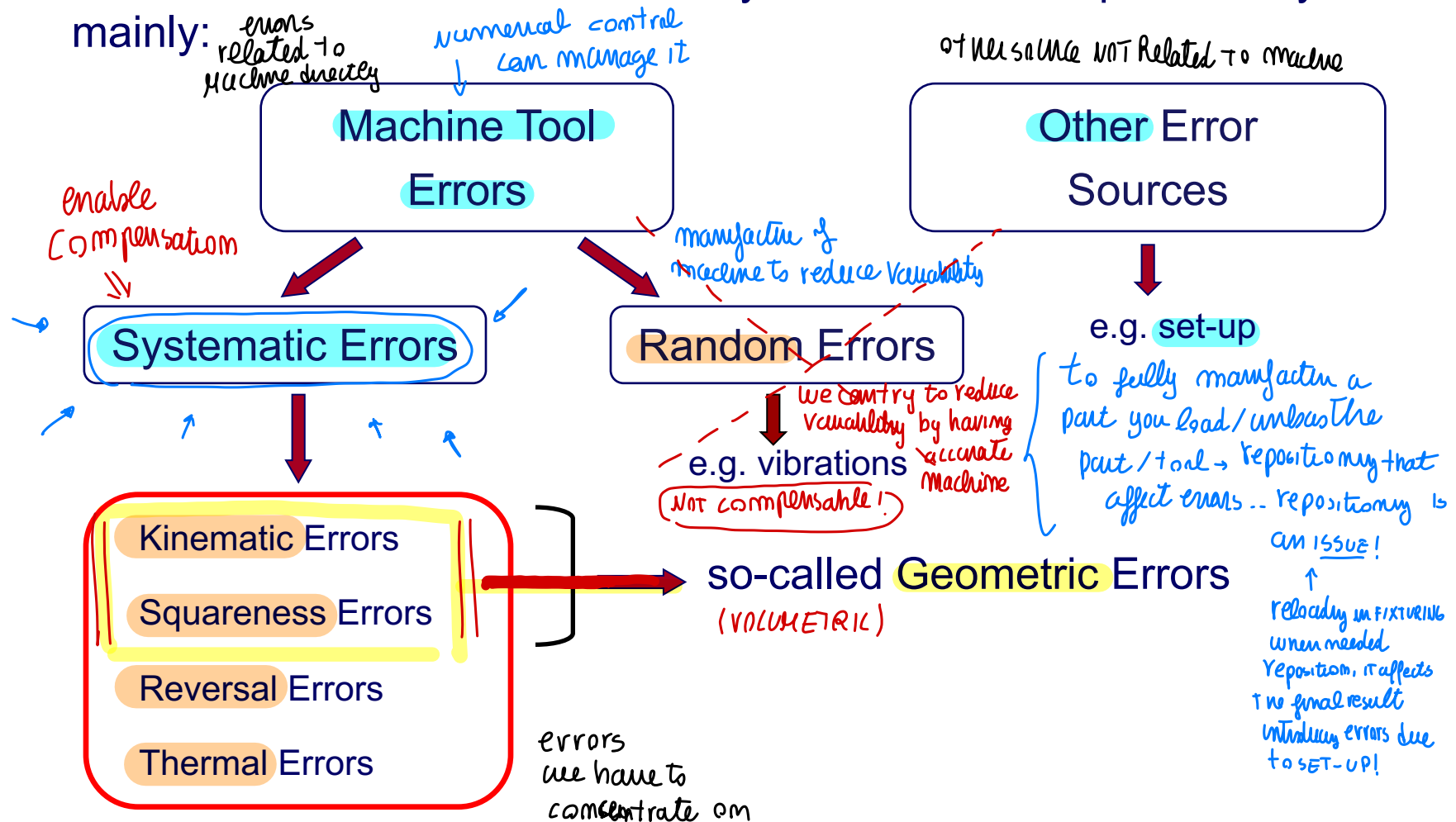


MACHINE TOOL ERROR SOURCES

different kind of errors!
"ERRORS" in CNC, CHM or machine tool

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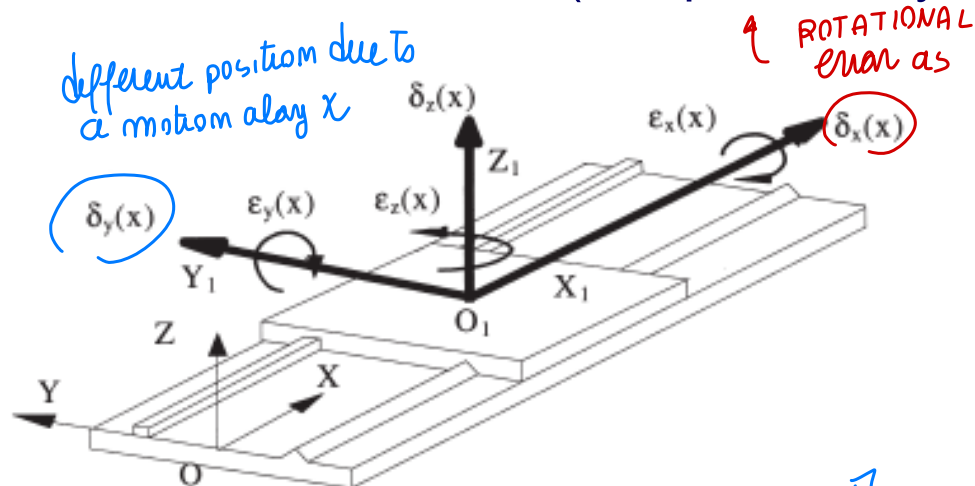
Machine tool performance and accuracy is affected by various error sources that influence the accuracy of the machined parts. They are mainly:





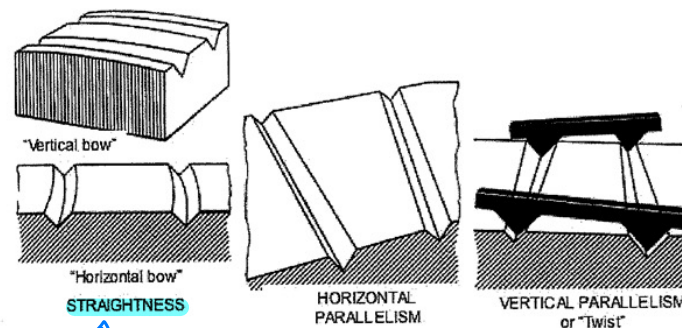
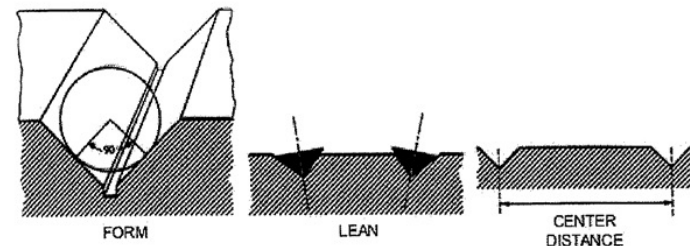
MACHINE TOOL ERROR SOURCES

Kinematic Errors: During the positioning of each axis it is possible to have up to six errors, three linear errors (a linear positioning error and two straightness errors) and three rotation errors (roll, pitch and yaw).



different position due to a motion along x

ROTATIONAL error as



When moving rigid bodies between, we have NOT perfect guideways → different kinds of errors
KINEMATIC

6 ERRORS because of axis!

along x, y, z due to motion

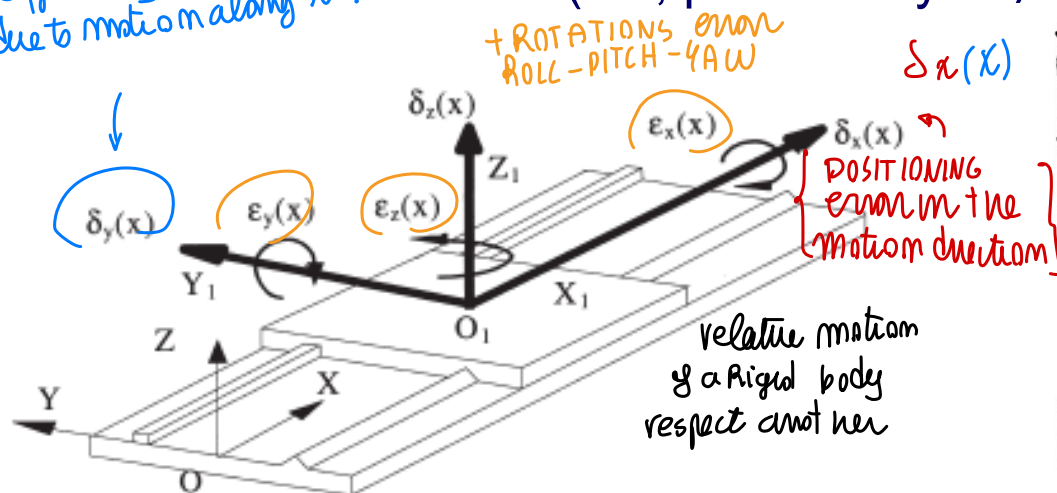
following a line with FORM error along x, y



MACHINE TOOL ERROR SOURCES

Kinematic Errors: During the positioning of each axis it is possible to have up to six errors, three linear errors (a linear positioning error and two straightness errors) and three rotation errors (roll, pitch and yaw).

different y position (or z) due to motion along x!



moving the part along the guideways → with errors!

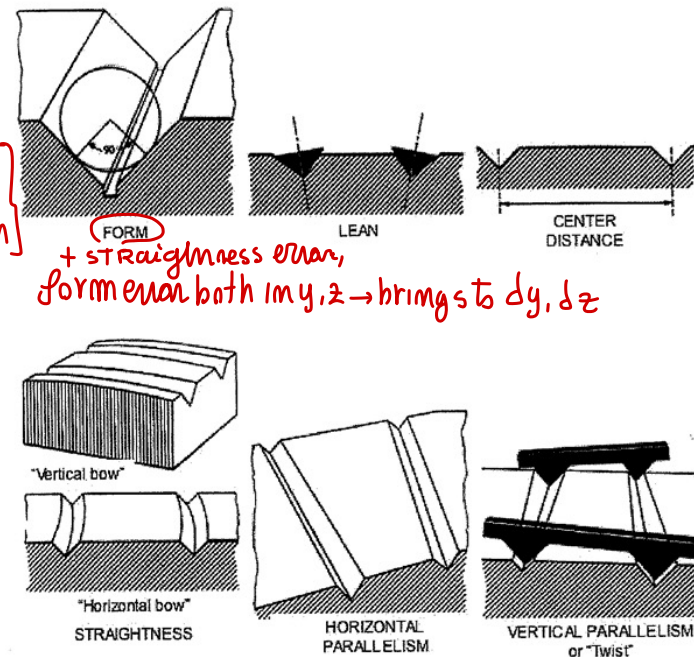
interaction between body - guideways → 6 different

kind of errors during that motion ⇒

$\delta_x, \delta_y, \delta_z, \epsilon_x, \epsilon_y, \epsilon_z$

6 errors / axis!

so on a MILLING machine { 6 × 3 axis = 18 errors }
(3 ORTHOGONAL AXIS)





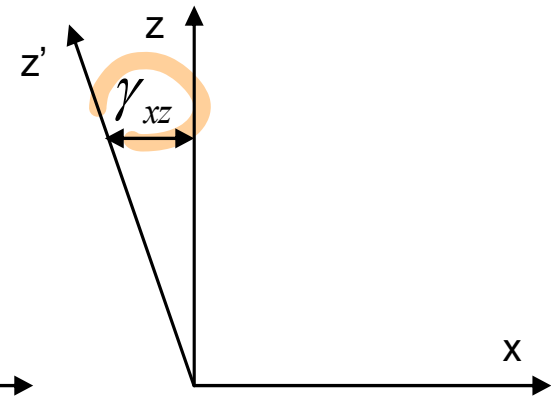
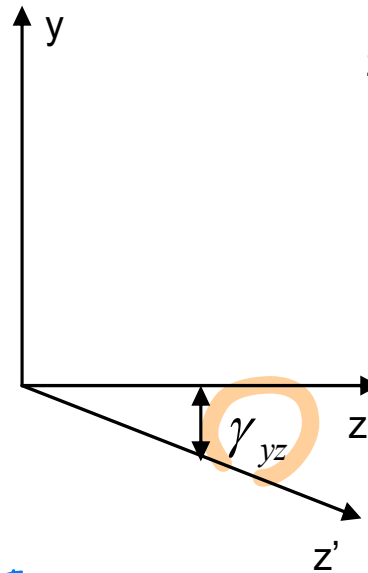
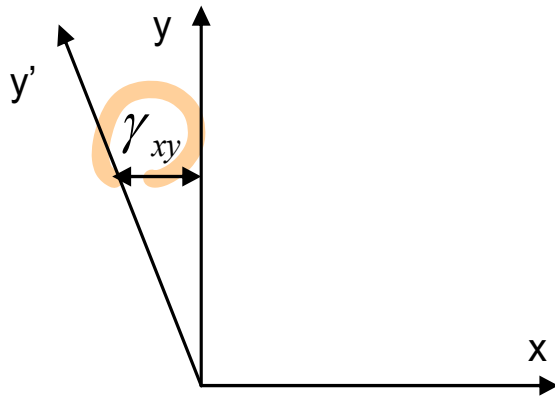
MACHINE TOOL ERROR SOURCES

ATTENTION on assembling of machine tool components

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connecting different axis you can have issue in the perpendicularity of axis (3 errors)

Squareness Errors: Perpendicularity errors between each couple of axes, mainly due to assembly.



{ Varying average ERRORS }

↳ geometrical error as combination of that

SQUARENESS error + KINEMATIC errors

Angular errors of the axis (perpendicularity error)



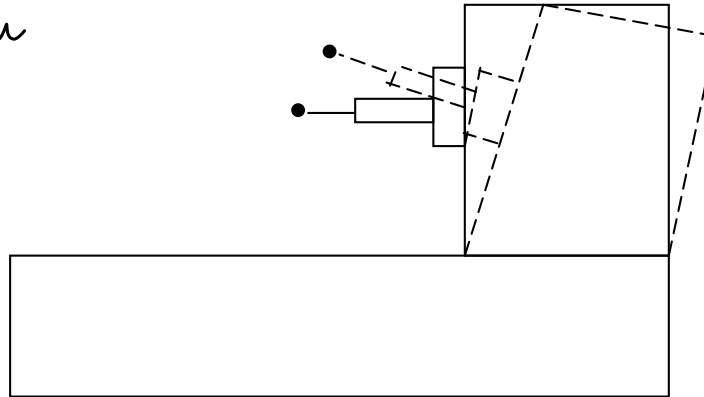
MACHINE TOOL ERROR SOURCES

Thermal Errors:

Structural deformation due to thermal effects.

Temperature effect creating **Deformations!** (high deform if overconstrained structure $\rightarrow$ highly deformy all)

in CMM we don't want overconstrained structure, but Aluminum free to move axis (subjected to linear deform that can be easily predicted and compensated)



Temperature deformation.. relevant for overconstrained system

(in CMM we prefer Aluminum axis NON Overconstrained to have a good prediction $\rightarrow$ easy compensation)



MACHINE TOOL ERROR SOURCES

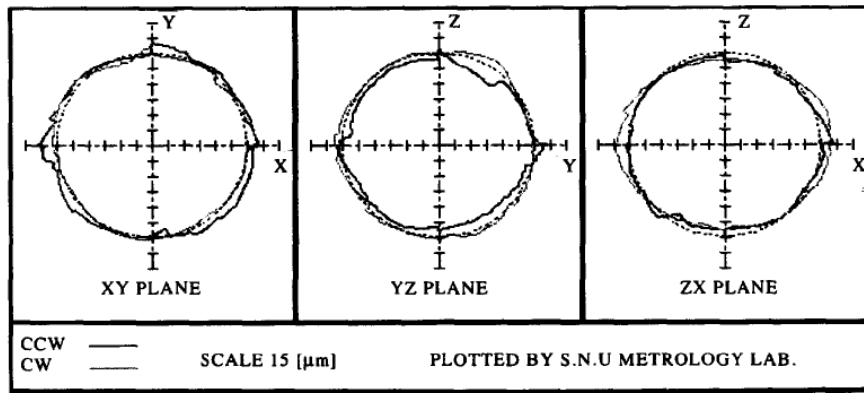
Related to axis error and interpolation, errors introduced when reversing one axis direction

ERRORS due also to control and transmission system

Reversal Errors:

related to interpolations when we reverse the axis

Errors due to many reasons during the change in the direction of the motion of a linear axis.

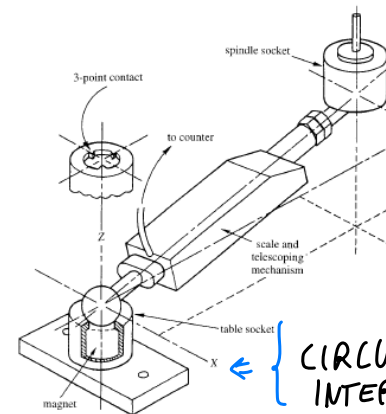


change axis motion direction

measure sensor position when it is moved on a plane -- CIRCULAR MOTION on a linear scale



Measuring device



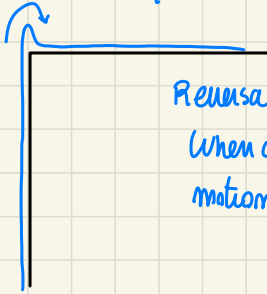
2D linear scale (measure sensor in a plane)

CIRCLE motions... ROTATING in different direction we have different ERRORS -- Multiple possible errors

CIRCULAR INTERPOLATION on Ref. plane

different motion than expected

different respect
a square



Reusnal error--

When changing kind of

motion $\Rightarrow$ 2D LINEAR scale

measuring linear position we
have measuring error during
angular interpolation

VOLUMETRIC ERROR *analyze that kind of errors!*

Kinematic, squareness, and thermal errors could be modeled, therefore they could be predicted, measured and compensated.

These models define the error of the positioning of the tool with respect to the workpiece inside the machine tool working envelope, the so-called volumetric error of the machine tool.

It is possible to consider different situations (hypotheses):

Machine tool as an assembly of rigid bodies

Machine tool as an assembly of quasi-rigid bodies

- Geometric Errors**

*look JUST to kinematics + squareness
NO POSSIBLE distortion*

← (different assumption) →

- Geometric Errors** *deformation due to KINEM + SQUARENESS*

- Thermal Errors** *(complex estimation)
analytical model hard to have*

We can have some deformation, also Thermal error

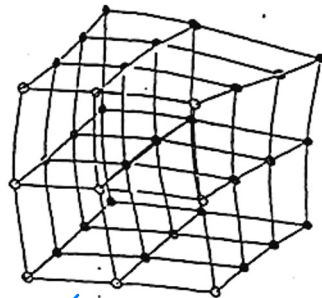
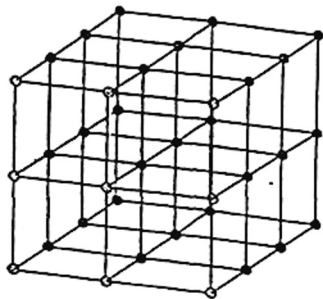


VOLUMETRIC ERROR

RIGID Bodies condition

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complex model to obtain



machine tool configuration

Respect to
kind of
val errors

	<u>3-axis</u> Machine Tool	5-axis Machine Tool
RIGID BODY <u>Geometric</u> Errors	3 squariness → 21-error model 6 × each axis = 18 → 3 + 18	going to 5 axis it depends on configuration the amount of error -- HIGH configuration dependent (>40 errors)
QUASI-RIGID BODY Geometric and Thermal Errors	configuration dependent (>40 errors)	very complex

immediately at the errors

↳ depending on config

⇒ considering a simple situation

↑
behaviour relevant for who manufacture that tools



VOLUMETRIC ERROR: 3-AXIS CNC MACHINE TOOL

Cartesian MT schema: **FXYZ**

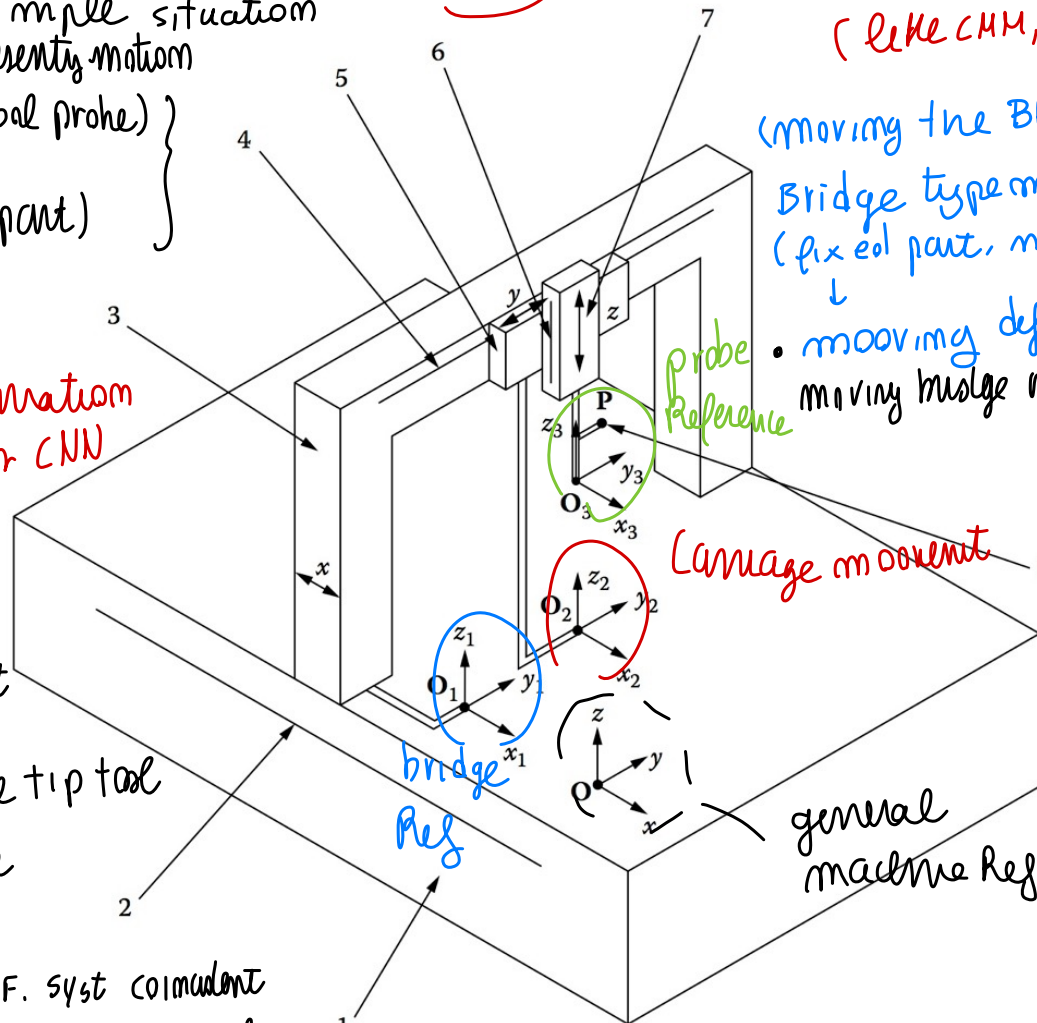
We are on that simple situation
MODEL of, when representing motion

$\left\{ \begin{array}{l} FXYZ \text{ (moving the tool/probe)} \\ \text{or} \\ XYZF \text{ (moving the part)} \end{array} \right\}$

configuration
used in CNC

We can
evaluate the (P)
position (PROBE) respect
the reference system
 $OXYZ$, knowing the tip tool
position respect sample

start from
Hyp. } all REF. syst coincident
in main Ref of machine $\rightarrow$



$\Downarrow$ FXYZ moving the probe,
became XYZ after "F"
(like CHM, probe move and part fixed)

(moving the BRIDGE)
Bridge type machine
(fixed part, moving the probe)

• moving different bodies
moving bridge respect base

Carriage movement

different Ref
syst with
parallel axis
but different
ORIGIN

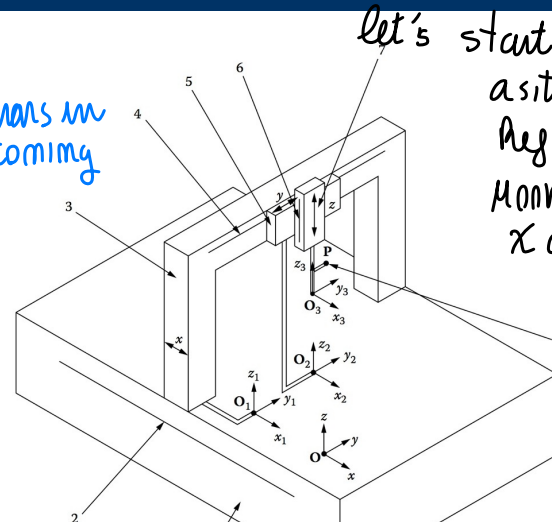


VOLUMETRIC ERROR: 3-AXIS CNC MACHINE TOOL

ROTATION Representation Respect
the axis through ROTARY MATRIX

$$OO_1 = X = \begin{pmatrix} \text{movement} \circledast X + \delta_x(X) \\ \delta_y(X) \\ \delta_z(X) \end{pmatrix}$$

linear errors in positioning



let's start to move probe from
a situation where all
the axes are coincident
moving the probe along the
X axis of a quantity X

$$(*) \begin{cases} \sin(\epsilon) \approx \epsilon \\ \cos(\epsilon) \approx 1 \end{cases}$$

+ ROTATIONS ERRORS (3 ROTARY MATRIX) effect of ROTATION depends on the center of rotation
IF big angle --- $\rightarrow$ here we have small errors angle! $\Rightarrow$ linearize

$$R(x) = \begin{pmatrix} \cos(\epsilon_z(X)) & \sin(\epsilon_z(X)) & 0 \\ -\sin(\epsilon_z(X)) & \cos(\epsilon_z(X)) & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos(\epsilon_y(X)) & 0 & -\sin(\epsilon_y(X)) \\ 0 & 1 & 0 \\ \sin(\epsilon_y(X)) & 0 & \cos(\epsilon_y(X)) \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\epsilon_x(X)) & \sin(\epsilon_x(X)) \\ 0 & -\sin(\epsilon_x(X)) & \cos(\epsilon_x(X)) \end{pmatrix}$$

$$R(x) \underset{(*)}{\approx} \begin{pmatrix} 1 & \epsilon_z(X) & 0 \\ -\epsilon_z(X) & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -\epsilon_y(X) \\ 0 & 1 & 0 \\ \epsilon_y(X) & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \epsilon_x(X) \\ 0 & -\epsilon_x(X) & 1 \end{pmatrix} \approx \begin{pmatrix} 1 & \epsilon_z(X) & -\epsilon_y(X) \\ -\epsilon_z(X) & 1 & \epsilon_x(X) \\ \epsilon_y(X) & -\epsilon_x(X) & 1 \end{pmatrix}$$

SMALL ANGLES
approx

no matter on the order of the ROTATION, full rotation can be represented
by this matrix! $\Rightarrow$ depends on 3 ROT. ERRORS



VOLUMETRIC ERROR: 3-AXIS CNC MACHINE TOOL

We can use homogeneous coordinates to represent ROTATION + TRANSLATION Errors!

$$\mathbf{OP} = \mathbf{P} = \begin{pmatrix} X_P & Y_P & Z_P \end{pmatrix}^T$$

$$\mathbf{O}_1\mathbf{P} = \mathbf{P}_1 = \begin{pmatrix} X_{P1} & Y_{P1} & Z_{P1} \end{pmatrix}^T$$

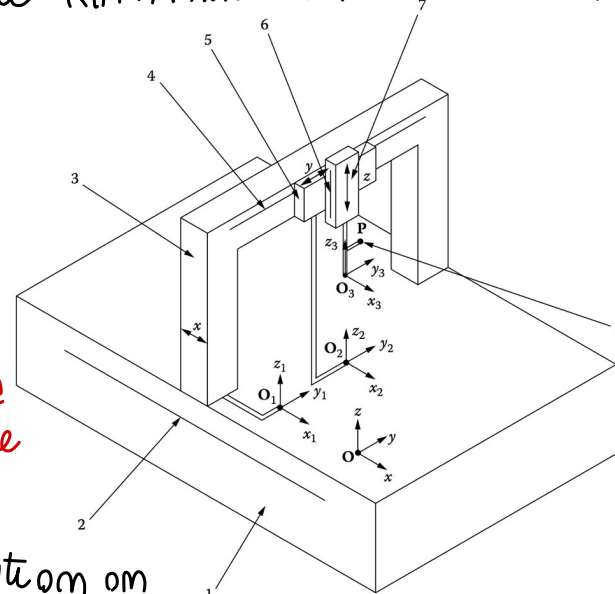
moving along X axis

→ in that way we compute the new probe position
respect first reference
of machine!

$$\mathbf{P} = \mathbf{X} + \mathbf{R}^{-1}(x)\mathbf{P}_1$$

Homogeneous coordinates to represent position on
Ref. plane of machine

$$\begin{pmatrix} X_P \\ Y_P \\ Z_P \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & \varepsilon_z(X) & -\varepsilon_y(X) & X + \delta_x(X) \\ -\varepsilon_z(X) & 1 & \varepsilon_x(X) & \delta_y(X) \\ \varepsilon_y(X) & -\varepsilon_x(X) & 1 & \delta_z(X) \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X_{P1} \\ Y_{P1} \\ Z_{P1} \\ 1 \end{pmatrix}$$





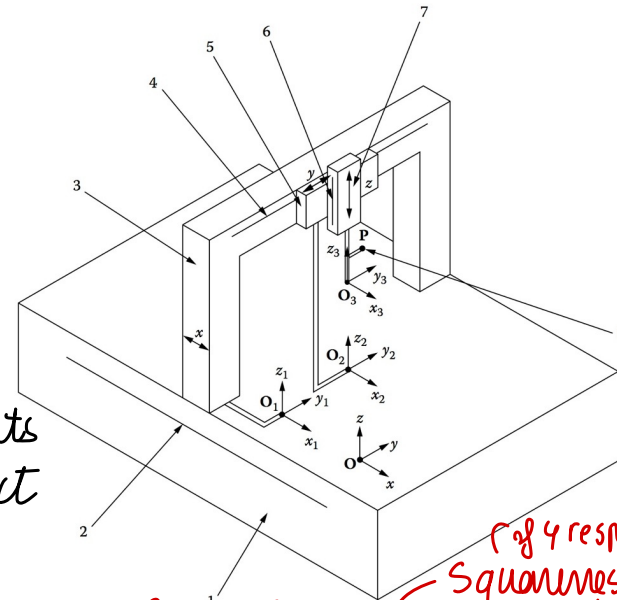
VOLUMETRIC ERROR: 3-AXIS CNC MACHINE TOOL

$$\mathbf{O}_1 \mathbf{P} = \mathbf{P} = \begin{pmatrix} X_{P1} & Y_{P1} & Z_{P1} \end{pmatrix}^T$$

$$\mathbf{O}_2 \mathbf{P} = \mathbf{P}_2 = \begin{pmatrix} X_{P2} & Y_{P2} & Z_{P2} \end{pmatrix}^T$$

IF moving y axis $\rightarrow$ similarly we have
Rotation components
+ linear error effect

$$\mathbf{P}_1 = \mathbf{Y} + \mathbf{R}^{-1}(y) \mathbf{P}_2$$



ROTATION

TRANSLATION

(of y respect x)
Squarness! to
be considered

$$\begin{pmatrix} X_{P1} \\ Y_{P1} \\ Z_{P1} \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & -\varepsilon_z(Y) & \varepsilon_y(Y) \\ \varepsilon_z(Y) & 1 & -\varepsilon_x(Y) \\ -\varepsilon_y(Y) & \varepsilon_x(Y) & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \delta_x(Y) - Y\alpha_{xy} \\ Y + \delta_y(Y) \\ \delta_z(Y) \\ 1 \end{pmatrix} \begin{pmatrix} X_{P2} \\ Y_{P2} \\ Z_{P2} \\ 1 \end{pmatrix}$$

motion of y along y axis -- now we can compute respect
machine position



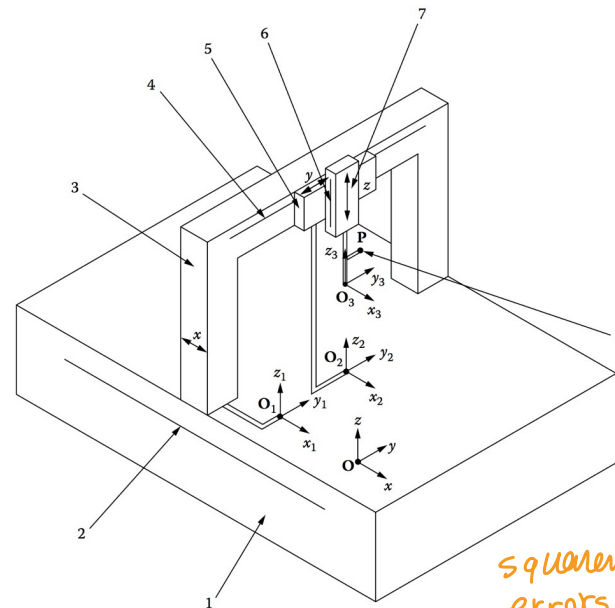
VOLUMETRIC ERROR: 3-AXIS CNC MACHINE TOOL

$$\mathbf{O}_2 \mathbf{P} = \mathbf{P} = \begin{pmatrix} X_{P2} & Y_{P2} & Z_{P2} \end{pmatrix}^T$$

$$\mathbf{O}_3 \mathbf{P} = \mathbf{P}_3 = \begin{pmatrix} X_{P3} & Y_{P3} & Z_{P3} \end{pmatrix}^T$$

moving at the end along z

$$\mathbf{P}_2 = \mathbf{Z} + \mathbf{R}^{-1}(z) \mathbf{P}_3$$



squariness errors effect due to along z motion

$$\begin{pmatrix} X_{P2} \\ Y_{P2} \\ Z_{P2} \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & -\varepsilon_z(Z) & \varepsilon_y(Z) & \delta_x(Z) - Z\alpha_{zx} \\ \varepsilon_z(Z) & 1 & -\varepsilon_x(Z) & \delta_y(Z) - Z\alpha_{yz} \\ -\varepsilon_y(Z) & \varepsilon_x(Z) & 1 & Z + \delta_z(Z) \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X_{P3} \\ Y_{P3} \\ Z_{P3} \\ 1 \end{pmatrix}$$

compute final position on $z \Rightarrow$



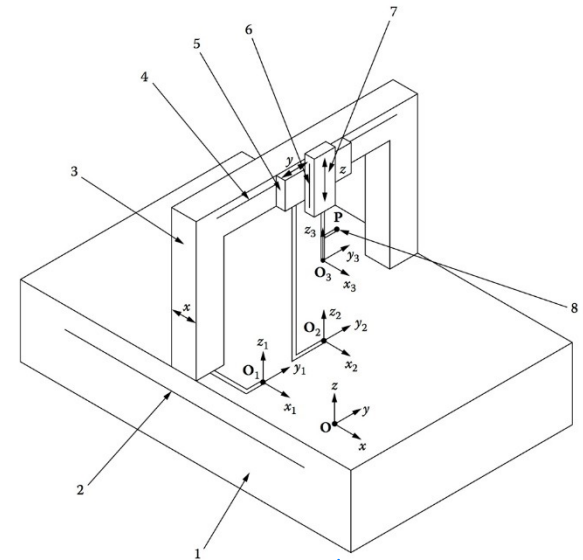
VOLUMETRIC ERROR: 3-AXIS CNC MACHINE TOOL

$$\mathbf{OP} = \mathbf{P} = \begin{pmatrix} X_P & Y_P & Z_P \end{pmatrix}^T$$

$$\mathbf{O}_3\mathbf{P} = \mathbf{P}_3 = \begin{pmatrix} X_{P3} & Y_{P3} & Z_{P3} \end{pmatrix}^T$$

$$\mathbf{P} = \mathbf{X} + \mathbf{R}^{-1}(x) \left\{ \mathbf{Y} + \mathbf{R}^{-1}(y) \left[\mathbf{Z} + \mathbf{R}^{-1}(z) \mathbf{P}_3 \right] \right\}$$

combining all the motions we can go from position respect probe head to the position of the TIP (tool point) respect machine Ref plane $\Rightarrow$



Ideal Machine Tool:

IF PERFECT machine $\rightarrow$ NO ERRORS!

$$\begin{cases} X_P = X + X_{P3} \\ Y_P = Y + Y_{P3} \\ Z_P = Z + Z_{P3} \end{cases} \quad \begin{array}{l} \text{JUST add} \\ \text{position} \\ \text{respect} \\ \text{3RD ref!} \end{array} \quad \neq$$

Real Machine Tool:

$$\begin{cases} \Delta X = X_P - (X + X_{P3}) \\ \Delta Y = Y_P - (Y + Y_{P3}) \\ \Delta Z = Z_P - (Z + Z_{P3}) \end{cases}$$

errors in positioning

But we have 21 errors model!

$\rightarrow$ analytically



The final result is:

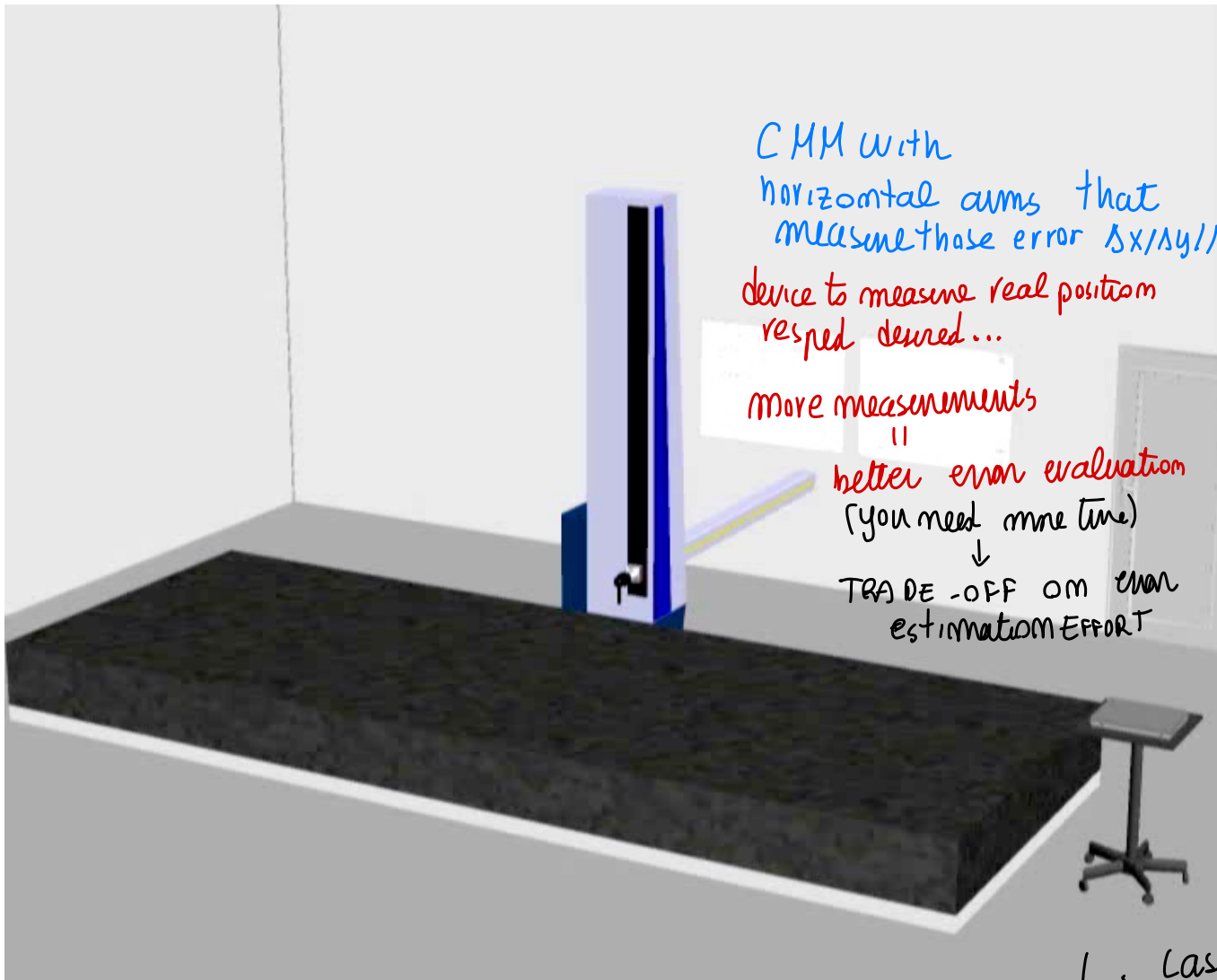
evaluated as functions of the possible ^{average errors} SYSTEMATICS errors as position function

$$\Delta X = \delta_x(X) + \delta_x(Y) + \delta_x(Z) - Y\alpha_{xy} - Z\alpha_{xz} - Y\varepsilon_z(X) + Z[\varepsilon_y(X) + \varepsilon_y(Y)] + \\ - Y_{P3}[\varepsilon_z(X) + \varepsilon_z(Y) + \varepsilon_z(Z)] + Z_{P3}[\varepsilon_y(X) + \varepsilon_y(Y) + \varepsilon_y(Z)]$$

$$\Delta Y = \delta_y(X) + \delta_y(Y) + \delta_y(Z) - Z\alpha_{yz} - Z[\varepsilon_x(X) + \varepsilon_x(Y)] + \\ + X_{P3}[\varepsilon_z(X) + \varepsilon_z(Y) + \varepsilon_z(Z)] - Z_{P3}[\varepsilon_x(X) + \varepsilon_x(Y) + \varepsilon_x(Z)]$$

$$\Delta Z = \delta_z(X) + \delta_z(Y) + \delta_z(Z) + Y\varepsilon_x(X) - X_{P3}[\varepsilon_y(X) + \varepsilon_y(Y) + \varepsilon_y(Z)] \\ + Y_{P3}[\varepsilon_x(X) + \varepsilon_x(Y) + \varepsilon_x(Z)]$$

↑ to predict the final
volumetric error along
 $X, Y, Z \Rightarrow \Delta X, \Delta Y, \Delta Z$
that so can be compensated



↳ Laser track
errors

<https://youtu.be/v4eb0hotUiw>



It adopts a laser interferometer to measure a distance, and simultaneously measures two angles, from which the position in space of the target is derived.



System
able to
measure in some
system
position / angles --

able to track the target -- such
the target in space $\rightarrow$ many of those sensors can be used to average the error

Generalize ERROR approach? $\Rightarrow$





iGPS (INDOOR GLOBAL POSITIONING SYSTEM)

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A constellation of emitters (passive) sends a signal and a sensor (active), by triangulation of these emitters, identifies its position.

If making big objects, to track positions and find errors, you need possible solution to use

(HIGH GPS) ↓

IN-DOOR GPS,
sensors tracking target position inside environments
⇒ always see the

target position! (constellation of sensors for tracking) → cellalogy
just also to track an AGV on the manufacturing plant





SOFTWARE VOLUMETRIC ERROR COMPENSATION

At present CNCs allows three possible representations of the model of the volumetric error :

- error grid;
 - error table;
 - coefficient table.
- (different representation of the errors measured)*
- Information that you can have in your control system*

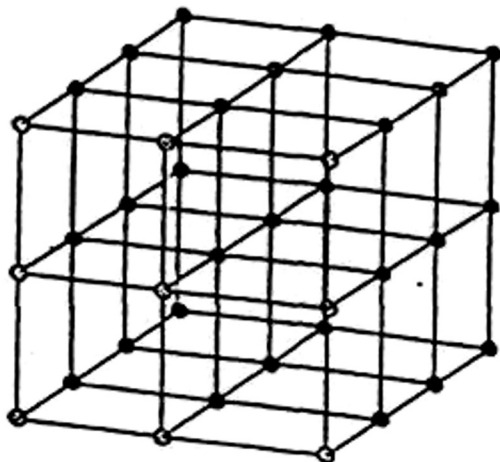


SOFTWARE VOLUMETRIC ERROR COMPENSATION

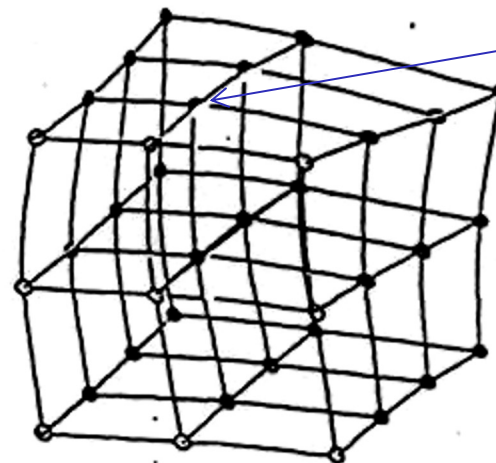
Error Grid: the working envelope of the machine tool is discretized by a regular grid, and for each node of the grid the three components of the error are memorized.

If we consider n point for each axis, we'll need $3n^3$ error components. Compensation in an intermediate point will be done by linear interpolation.

NOT so used in CNC machine tools $\Rightarrow$



n ↑ is better knowledge of the error



$$\begin{cases} \Delta X_i \\ \Delta Y_i \\ \Delta Z_i \end{cases}$$

DISCRETIZATION
of volumetric error

↓
to have continuous $\Delta X, \Delta Y, \Delta Z$
YOU INTERPOLATE!

keep information about $\Delta X, \Delta Y, \Delta Z$ $\forall$ point $\rightarrow 3n^3$ INFO

$\rightarrow$ total n^3 POINTS $\rightarrow$ and having $\Delta X, \Delta Y, \Delta Z$ $\forall$ points

↓
you have
ALL ERROR
INFO



SOFTWARE VOLUMETRIC ERROR COMPENSATION

not used in CNC machine tools

Error Table: For each axis, discretised in n points, we keep memory of the corresponding value of each kinematic error function. The value of a kinematic error function in an intermediate point of the axis will be computed by linear interpolation.

POINT, discretized $\rightarrow$ you need Δm
interpol. to continuous

Starting from these data, the volumetric error is computed from its model.

(you have the 3 values of squareness error, not depend on position (3 squares))
 $\downarrow$ here you represent the other 18 errors ($F_1, \dots, F_{18}$) as function of position on axis

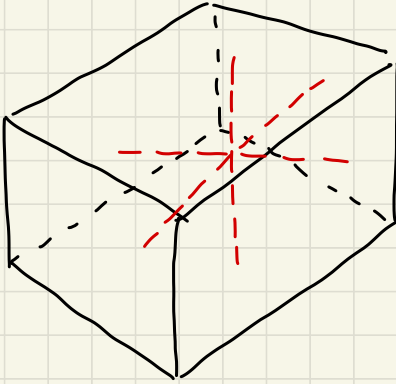
X Axis	Kinematic errors as function of axis position			
Position	F1	F2	...	F18
1	...			
2				
3				
...				
n-1				
n	discretizing the axis			...

table of
X axis
you will
have in
total 3
tables
X, Y, Z axis

knows errors
respect axis
position
with
 $\Delta X, \Delta Y, \Delta Z$
positional error
in different point,
by expressing the
FUNCTION

IF you have one table & axis... there is an issue...

If you have a certain **WORKING ENVELOPE**



discretized motion for
the 3 axis

moving machine along
X axis staying in
the middle of the other two
axis...

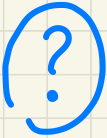
but if positioning another
point of working envelop...
we don't know how..

LIMIT! measurement

done JUST along each

axis → no idea of how different axis are correlated --
influencing in between

you can have an average on the table error but unknown
motion in the middle → LIMIT!



NON SI
CAPISCE
NA MINCHIA



SOFTWARE VOLUMETRIC ERROR COMPENSATION

IF instead of the function table we fit a model $\rightarrow$ error model fitting (18 functions polynomial of position)

Coefficient Table: If we know the model of each kinematic error function, it is possible to memorize only the coefficients of the model. Usually the model could be an interpolating cube polynomial function.

order 3 is enough for error

$$f1: a_1 + b_1x + c_1x^2 + d_1x^3$$

$$f2: a_2 + b_2x + c_2x^2 + d_2x^3$$

.....

$$f18: a_{18} + b_{18}x + c_{18}x^2 + d_{18}x^3$$

problem of interconnectivity
multiple axis as before, BUT
better model, better performances

$$\longrightarrow [18 \cdot (o + 1)]$$

Memorized data

{ RARE situation }

Instead of discretize n values, we try to evaluate
 $\rightarrow$ better INFORMATION, NOT JUST
interpolating but error model fitting
data

4 coefficient $\forall$ error
IF 3RD ORD. polynomial



SOFTWARE VOLUMETRIC ERROR COMPENSATION

ERROR REPRESENT
IN CNC machine tool

Error Grid ← need a lot of data!

- Allows a complete representation of the volumetric error in the whole measuring volume of the machine.
- Only few CNCs allow it.

Error Table ← MOST USED

- Very diffused, as it is coherent with the commonly adopted methods for the measurements of the volumetric error.
- Difficulties in the identification of the interactions between the various axes.

Coefficient Table ← used by FEW CNC machine, BUT it does NOT affect on considering multiple axis at one time!

- Potentially effective.
- Very complicated (hard to manage with current CNCs).



VOLUMETRIC ERROR COMPENSATION

WAY TO COMPENSATE the errors

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Once the volumetric error is known, there are three ways to consider it to improve the precision of the machine tool:

CNC OR CMM

1. **Hardware correction** (components reworking); *iteratively a lot used, when measure error we disassemble machine and rework it to reduce errors by physically modify the machine*
(only PAST method possible!) → RE ASSEMBLE until LOW ERROR!
2. **Software compensation of measured position** (open loop compensation, it applies only to CMS): *ONLY CMM*

after positioning, we compensate it, applied only on CMM (Goal to reach the point even if small error → compensate that error)

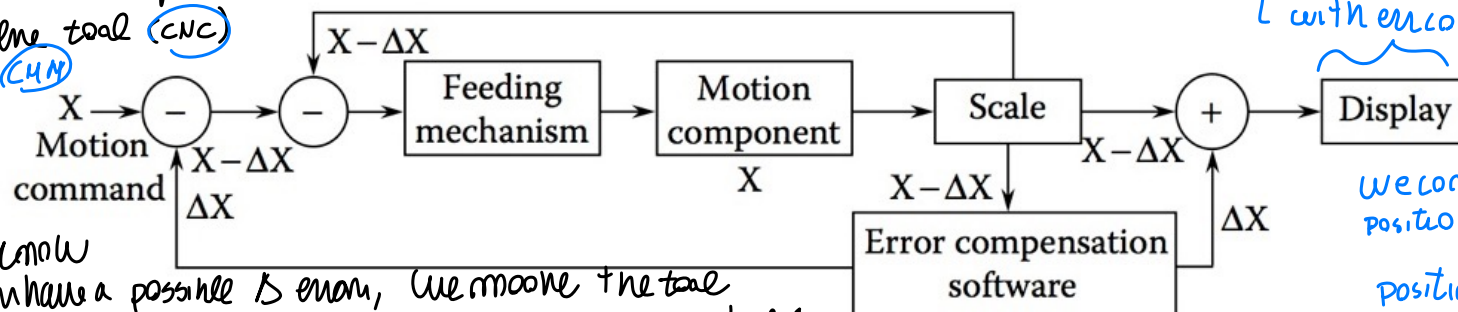
$$\begin{pmatrix} X + \Delta X; & Y + \Delta Y; & Z + \Delta Z \end{pmatrix}$$

correct sample point using $\Delta x, \Delta y, \Delta z$ THIS

cannot be done on Machine tool (error in machine is NOT admissible!)

3. **Software volumetric error compensation according to the following schema** (closed loop compensation):

perform error comp. in machine tool (CNC) and also for CMM



read RIGHT POSITION!
I see the right real value with error compensation

AVOID MISTAKE

We control positioning

positioning control loop + external error comp. control loop

If we know we can have a possible Δ error, we move the tool compensating the error on closed-loop (external control loop) respect axis control loop compensation depending on the space, knowing the error!